

Basic Principles of Medicine 2

Nervous System and Behavioral Science

Neurotransmitter Pathways and Roles

DLA (Part 2)

Learning Objectives

- SOM.MK.I.BPM2.1.NB.1.NSCI.0109 Describe the synthesis/removal of neuroactive peptides
- SOM.MK.I.BPM2.1.NB.1.NSCI.0110 Describe the synthesis/removal of nitric oxide

Neuroactive Peptides: Beta-endorphin

Synthesis

Originates from the pre-propeptide pre-proopiomelanocortin in the rough endoplasmic reticulum of the anterior and intermediate pituitary and the arcuate hypothalamic nucleus

b. Within the rough endoplasmic reticulum, the propeptide proopiomelanocortin arises

c. **Proopiomelanocortin (POMC)** undergoes packaging in the smooth endoplasmic reticulum

d. Fast orthograde transport drives vesicles from the hypothalamus to the periaqueductal grey and noradrenergic nuclei

e. **Proteolytic cleaving yields beta-endorphin, among other peptides**

Removal

Degradation by peptidases

Function and Distribution

Inhibitory at most synapses in the medulla and spinal cord

Opioid peptides are important participants in central pain-regulating pathways.

Neurotransmitter	Ionotropic Receptor	Metabotropic Receptor
Opioid peptides	—	Mu, delta, kappa, ORL1

Neuroactive Peptides: Enkephalins

Synthesis

- a. Originate from the pre-propeptide **pre-proenkephalin** in the rough endoplasmic reticulum of spinal and caudal bulbar neurons
- b. The resultant proenkephalin undergoes packaging and fast orthograde transport
- c. **Proteolysis yields leucine - and methionine-enkephalin, which are then loaded into dense-core vesicles in preparation for exocytosis**

Removal

Degradation by peptidases

Function and Distribution

Inhibitory at most synapses in the medulla and spinal cord

Neurotransmitter

Opioid peptides

Ionotropic Receptor

—

Metabotropic Receptor

Mu, delta, kappa, ORL₁

ORL: opioid **receptor** like-1

Substance P

Synthesis

An 11-amino acid peptide(tachykinin family) synthesized from a large precursor protein in the endoplasmic reticulum and transported to the nerve terminal

Removal

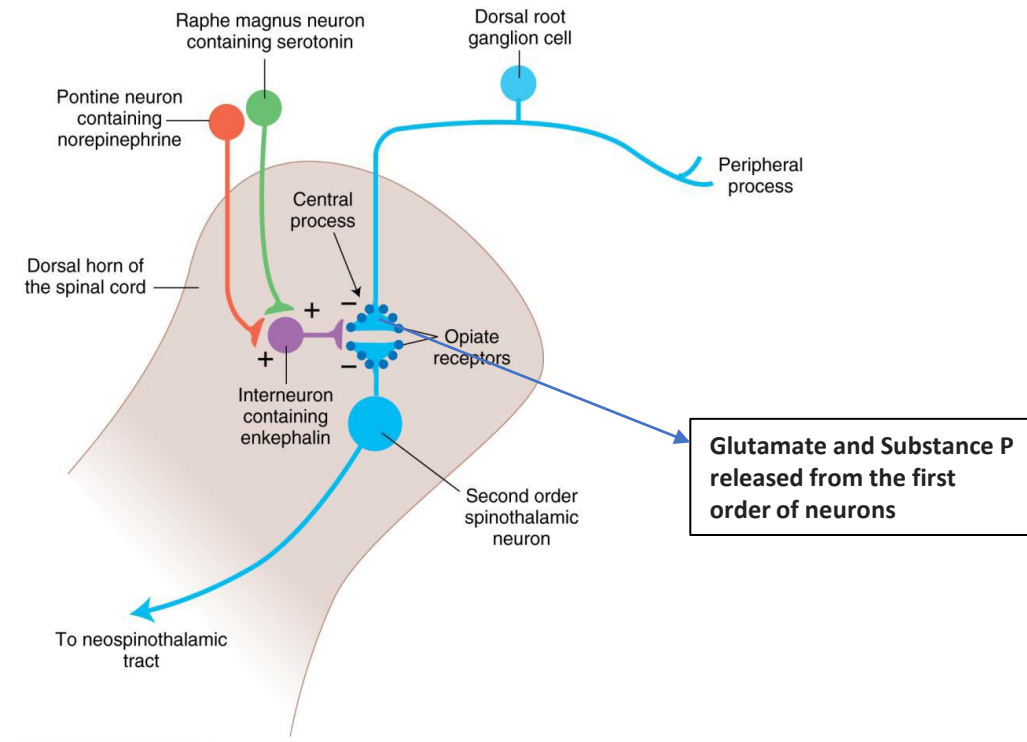
Degradation by peptidases

Function and Distribution

Excitatory at some synapses and inhibitory at other synapses; widely distributed throughout the CNS

Substance P: Sources and Interactions

- Synthesized in peripheral unmyelinated nociceptive fibers
- Released into spinal dorsal horn and spinal nucleus of the trigeminal nerve
- Inhibited indirectly by 5-HT and NE, which excite enkephalinergic neurons



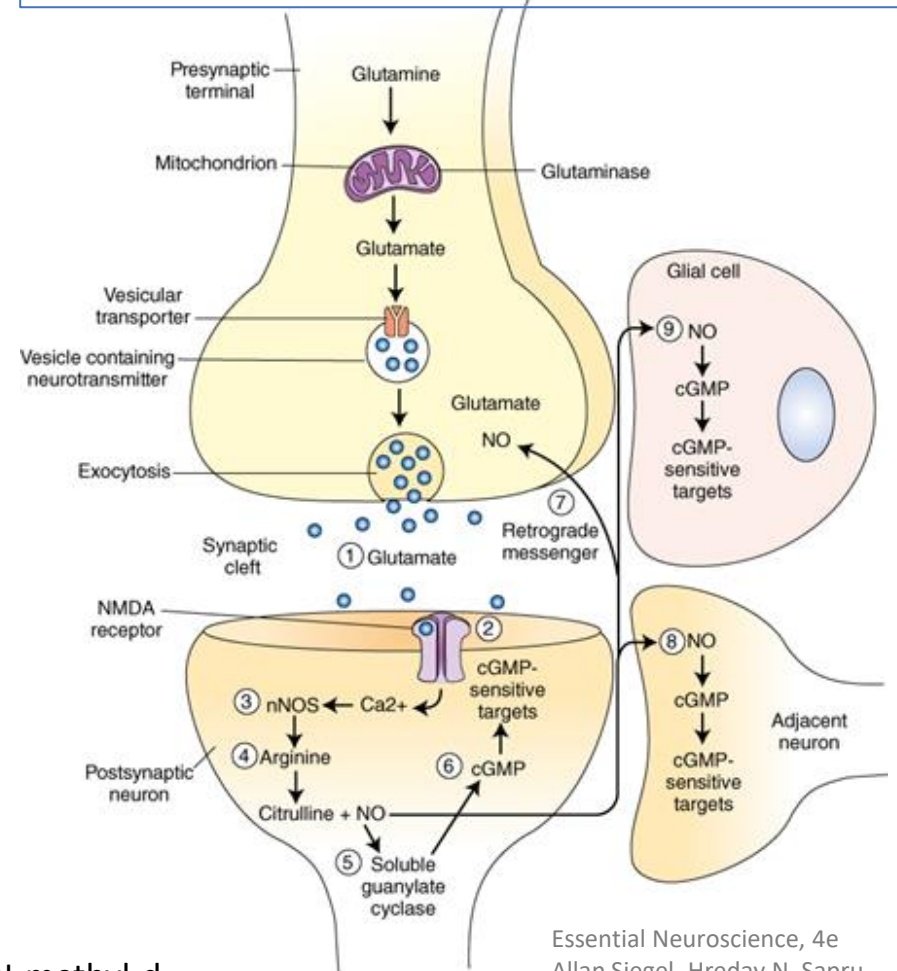
Gaseous Neurotransmitters: Nitric oxide (NO)

General Principles

Nitric oxide (featured below) and **carbon monoxide** have been identified as diffusible intercellular messengers. They are thus considered as neurotransmitters, despite their **failure to meet traditional criteria for transmitters**.

1. Not stored in vesicles
2. Non-exocytotic
3. Inactivation is passive and spontaneous
4. No surface receptors
5. Retrograde transmission possible

Steps involved in the synthesis of nitric oxide at a glutamatergic synapse



Ca²⁺, calcium; cGMP, cyclic guanosine monophosphate; NMDA, N-methyl-D-aspartic acid; nNOS, neuronal nitric oxide synthase isoform I; NO, nitric oxide.

Essential Neuroscience, 4e
Allan Siegel, Hreday N. Sapru,
Chapter 7: Neurotransmitters

SOM.MK.I.BPM2.1.NB.1.NSCI.0110

Nitric Oxide

Synthesis and Removal

The mechanism for the synthesis of nitric oxide (NO) varies among tissues. Some neurons express a neuron-specific isoform of **nitric oxide synthase** (nNOS), which triggers synthesis, as follows:

1. Glutamate activates post-synaptic NMDA receptors
2. Ca^{++} entering through the NMDA-related pore combines with calmodulin to activate Nnos
3. nNOS interacts with L-arginine to yield NO and L-citrulline
4. NO activates guanylate cyclase, promoting synthesis of cGMP in either the NO-expressing cell itself or, after intercellular diffusion, neighboring cells (neurons or glia)

Physiological and Clinical Considerations

•**Neurotransmission:** NO plays important role in memory formation by increasing the synaptic strength in the brain.

•**Vasodilation and Increase in Blood Flow:** sildenafil (Viagra), is used to treat male erectile dysfunction by increasing the level of cGMP, leading to increased blood flow to the penis.

•**Immune System:** NO causes immune responses. NO can cause apoptosis in cells, resulting in their death. NO may also promote inflammation.

•**Role in Multiple Sclerosis (MS):** in MS, NO disrupts the blood–brain barrier, causes oligodendrocyte injury and demyelination, induces axonal degeneration, and that it contributes to the loss of function by impairment of axonal conduction.

•**Role in Alzheimer's Disease (AD):** Up-regulation of NOS expression and overproduction of NO in neurons and glia → Dysfunction of normal NO-mediated function in the cerebral vasculature and NO-mediated neuroinflammation and neurotoxicity → may contribute to the pathology of AD.

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